FIKR Model — Lesson Plan

Learning Leading to Production

Title	The Quadratic Formula	**Subject / Unit**	Topic 2 — Quadratic Functions and Equations · Lesson 2-6
Grade	Grade 10 — Algebra II	**Date**	Week 8 · Semester 1, 2026–27
Format	[ ] 40-minute [X] 60-minute	**Teacher**	Mr Malek Thiab

Essential Question (curriculum map)

Not shown — the Scope and Sequence document has no Essential Question column, so it is left open rather than invented. Add it when confirmed.

Lesson Objectives (curriculum map — verbatim)

1. Analyze the discriminant to identify the number and type of solutions for a quadratic equation.
2. Solve quadratic equations using the quadratic formula, including those with nonreal solutions.

Standards (curriculum map — verbatim)

HSN.CN.C.7 Solve quadratic equations with real coefficients that have complex solutions.

HSA.REI.B.4.A Use the method of completing the square to transform any quadratic equation in x into an equation of the form $(x - p)^2 = q$ that has the same solutions. Derive the quadratic formula from this form.

HSA.REI.B.4.B Solve quadratic equations by inspection (e.g., for $x^2 = 49$), taking square roots, completing the square, the quadratic formula and factoring, as appropriate to the initial form of the equation. Recognize when the quadratic formula gives complex solutions and write them as a ± bi for real numbers a and b.

Vocabulary (curriculum map) · Assessments (curriculum map)

Vocabulary: Discriminant ; quadratic formula.

Assessments: “Give it a go” questions during practice ; “Time to Check” at the Mastery Gate.

0 Preparation — Before Class

Learning Outcome: What will the student be able to do?

Students will find the discriminant $b^2 - 4ac$ and use it to state the number and type of solutions of a quadratic equation, including when it contains an unknown k (objective 1; HSA.REI.B.4.B).

Students will solve quadratic equations using the quadratic formula, with exact answers (objective 2; HSA.REI.B.4.B and HSA.REI.B.4.A).

Students will solve quadratic equations with nonreal solutions and write them in the form $a \pm bi$ (objective 2; HSN.CN.C.7 and HSA.REI.B.4.B).

Three teaching days this week. If time runs short, cut Smart Production and carry it into the next session —

never the Mastery Gate, which is the evidence.

Pre-Class Preparation — Topic/Link and Evidence of Readiness: What will the student complete before class?

Flipped video (4 min): "Where does the formula come from?" — completing the square on $ax^2 + bx + c = 0$ one step at a time. Posted on the LMS 48 hours ahead.

Two embedded EdPuzzle questions: complete the square on $x^2 + 6x$, and write $\sqrt{-16}$ using i.

Evidence of readiness: EdPuzzle completion report plus both embedded answers submitted before the bell.

1 Diagnostic 6 min

Tool Used: What evidence indicates differences in students' current levels?

3-question Quizizz (≤ 4 minutes) — the slide-4 checks: evaluate $b^2 - 4ac$ for $a = 2,\ b = -3,\ c = -5$; simplify $\frac{4 + \sqrt{20}}{2}$; write $\sqrt{-36}$ using i.

Answers: 49 ; $2 + \sqrt{5}$; $6i$. No calculator.

Where is the learning gap? What will you adjust in your instruction accordingly?

Expected gap 1 — Q1: $(-3)^2$ is written as -9, or the sign of $-4ac$ is lost; the discriminant is then wrong before the formula starts.

Expected gap 2 — Q2: only one term of the numerator is divided by 2.

Routing: 0–1 correct → re-teach substituting negative numbers and dividing a whole numerator. 2 → straight to the formula. 3 → move quickly through block 1 and begin the Investigate prompt early.

2 Targeted Instruction 18 min

Required Instruction and Modeling: What needs direct explanation? Which example will demonstrate the thinking process and the expected quality standard?

Slide 5 — The quadratic formula (objective 2; HSA.REI.B.4.A). $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Examples: $x^2 - 5x + 6 = 0$ gives $x = 3$ or 2; $2x^2 + 3x - 5 = 0$ gives $x = 1$ or $-\frac{5}{2}$; $x^2 - 4x - 1 = 0$ gives $x = 2 \pm \sqrt{5}$.

Slide 6 — The discriminant (objective 1; HSA.REI.B.4.B). $\Delta = b^2 - 4ac$. Examples: $x^2 + 6x + 9$ has $\Delta = 0$; $x^2 + x - 3$ has $\Delta = 13$; $2x^2 - 3x + 5$ has $\Delta = -31$.

Slide 7 — Nonreal solutions (objective 2; HSN.CN.C.7). Examples: $x^2 - 4x + 13 = 0$ gives $x = 2 \pm 3i$; $2x^2 + 2x + 5 = 0$ gives $x = -\frac{1}{2} \pm \frac{3}{2}i$; $x^2 + 2x + 2 = 0$ gives $x = -1 \pm i$.

Slide 8 — The discriminant with an unknown (objective 1). $x^2 - 4x + k = 0$ has one real solution when $k = 4$; $x^2 + kx + 16 = 0$ when $k = 8$ or $k = -8$.

Narration focus — find the discriminant first: its sign tells you what kind of answer to expect.

Quick Check for Understanding: What question or evidence confirms that students are ready to move on to practice?

On whiteboards, 60 seconds: find the discriminant of $x^2 + 4x + 3$ and of $x^2 + 4x + 5$.

Expected: 4 and -4. Watch for 16 (only b^2 found) and 28 (the sign of c lost).

80% correct → release guided practice. Below 80% → one more modelled example with a different sign pattern.

3 Practice 12 min

Guided Task: How will the student progress from guided to independent practice? What feedback will the student use?

Two "Give it a go" problems, identical for every student, worked together with the teacher:

(1) Solve $2x^2 - 5x - 3 = 0$. (2) Find Δ for $x^2 - 6x + 10 = 0$ and classify the solutions.

Answers: (1) $\Delta = 49$, so $x = 3$ or $x = -\frac{1}{2}$. (2) $\Delta = -4 < 0$, so two nonreal solutions.

Feedback: worked live on the board step by step; students self-mark and circle their OWN first error.

Independent Task and Feedback Tool: students choose their route and may switch at any time

Delivered through the "Choose Your Route" sheet with live feedback in Pear Deck. Routes are named for the task, not for the student.

PRACTICE — Secure the method. Find the discriminant of three equations and say how many real solutions each has; solve $x^2 + 5x + 3 = 0$, $2x^2 - 5x - 3 = 0$ and $x^2 - 2x + 10 = 0$. Done when: a, b and c are written with their signs and the discriminant is found first.

APPLY — Use it in context. Solve $2x^2 + x - 4 = 0$ exactly, decide which Jeddah Season prices give a model revenue of 3 000 SAR, and find the values of b for which $2x^2 + bx + 8 = 0$ has one solution. Done when: the discriminant decides the number of answers before any solving, and units are stated.

INVESTIGATE — Find out why. Derive the formula by completing the square; show that a perfect-square discriminant gives rational solutions; explain why nonreal solutions come in conjugate pairs. Done when: your argument names the discriminant and keeps the equation balanced at every step.

Every student sits the same Mastery Gate. The gate is not differentiated, because it is the evidence.

4 Production — Stage One 12 min

Production Task: What new task, situation, or product will allow the student to demonstrate learning?

Non-routine transfer task — this context has not appeared in practice (slide 10).

Organisers of a Jeddah Season concert, part of the Vision 2030 entertainment programme, model the ticket revenue as $R(p) = -10p^2 + 400p$ riyals when each ticket costs p riyals. The model values are invented for practice.

(a) Find the prices that give $2\,000$ SAR. (b) Show that $4\,000$ SAR is possible at exactly one price. (c) Use the discriminant to explain why $5\,000$ SAR cannot be reached.

Answers: (a) $\Delta = 800$, so $p = 20 \pm 10\sqrt{2}$, about 34.14 or 5.86 SAR. (b) $\Delta = 0$, so $p = 20$ SAR. (c) $\Delta = -400 < 0$: no real price.

Students then hand their reasoning to the AI critic.

Product Quality Criteria and the Role of AI: What is the minimum acceptable standard? What must the student complete independently?

Minimum acceptable standard: each task rewritten as a quadratic equal to zero with a, b, c stated, the discriminant calculated, its sign linked to the number of prices, and units (SAR) in every answer.

Role of AI: critic only. Briefed prompt — "Challenge my reasoning and point out any step I have not justified." Never "solve this".

Completed independently by the student: the written justification for parts (a)–(c). AI may challenge it; it may not write it.

5 Presentation — Evidence of Learning 6 min

Evidence of Learning and Success Criterion: What will the student present or explain, and how? What must be clearly demonstrated for learning to be considered achieved?

"Time to Check" — one individual task. No notes, no partner, no AI.

1. How many real solutions does $3x^2 - 4x + 5 = 0$ have? 2. Solve $x^2 + 6x + 4 = 0$. 3. Solve $x^2 + 2x + 17 = 0$.

Answers: 1. $\Delta = -44 < 0$: none (two nonreal). 2. $\Delta = 20$, so $x = -3 \pm \sqrt{5}$. 3. $\Delta = -64$, so $x = -1 \pm 4i$.

Done when: question 1 finds the discriminant before answering and question 3 is written in the form $a \pm bi$.

Scored live in Pear Deck as submissions arrive; result logged in the Impact Dashboard Mastery Rate.

6 Final Product — Refinement 6 min

Refinement and Final Product: What revision will the student make based on feedback? What will the final product look like?

Students revise their ticket-price reasoning using the AI critique and the teacher's micro-conference note.

Final product, three equally acceptable formats: a Canva one-pager, a photographed hand-written solution, or a 45-second voice note.

Required content: the three discriminants and what each says; the two prices for $2\,000$ SAR; ONE worked non-example — using $b = -400$ as $+400$ in the formula — with a sentence on how the sign changes the answer; and a caption on why $5\,000$ SAR is impossible.

Graded on originality and clarity of reasoning, not correctness alone.

Presentation/Publication and Reflection: Where will the product be presented or published? What reflection question will close the learning cycle?

Published to the class LMS portfolio; the strongest go on the Mathematics Department wall as the Topic 2 modelling set.

Reflection question: "Which is harder for you — finding the discriminant, or keeping the signs right in the

formula?”

Total minutes across the seven stages: **60** (6 + 18 + 12 + 12 + 6 + 6) — matches the 60-minute format, including transitions.

★ Required Pillars

Differentiation · Adaptive Learning · Saudi Connection · Exam Alignment

Differentiation — three routes named for the task; students choose, and may switch at any time

PRACTICE — Secure the method. Discriminants, then three equations to solve. Done when: a, b, c have their signs and the discriminant comes first.

APPLY — Use it in context. An exact solution, a Jeddah Season revenue and an unknown b. Done when: the discriminant decides the number of answers before solving.

INVESTIGATE — Find out why. The formula, rational solutions and conjugate pairs. Done when: the argument names the discriminant and stays balanced.

No student is labelled. The routes describe tasks; every student sits the same Mastery Gate.

Adaptive Learning — how the lesson branches on evidence

Entry branch — the Quizizz result routes each student to the re-teach table, straight to the new content, or the Investigate prompt.

Mid-lesson branch — Pear Deck flags error patterns during independent practice; flagged students get a 2-minute micro-conference while others continue.

Exit branch — the Mastery Gate routes each student: PASS → Enrichment & Challenge; NOT YET → Targeted Learning Clinic at the start of the next lesson.

Routing is communicated privately through the LMS, never read aloud.

Saudi Culture & Real-World Connection

Production and project context: a Jeddah Season concert in a Vision 2030 entertainment setting, with model ticket prices in SAR (invented values).

Discussion prompt: why does a zero or negative discriminant tell the organisers that a revenue target cannot be reached at more than one price, or at any price?

SAT / SAAT (Tahsili) / GAT (Qudurat) Connection

SAT — Advanced Math: $x^2 - 8x + c = 0$ has exactly one real solution when $c = 16$. Trap: $c = 4$ (half of b) or $c = 64$ (forgetting the $4a$).

GAT (Qudurat) — which is greater, the number of real solutions of $x^2 + 4x + 7 = 0$ or 1? $\Delta = -12$, so $0 < 1$: B. Trap: assuming every quadratic has two solutions.

SAAT (Tahsili) — the solutions of $2x^2 - 3x - 1 = 0$ are $\frac{3 \pm \sqrt{17}}{4}$. Trap: dividing by 2 instead of $2a = 4$.

Suggestions for Improvement: